

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: RAYAVARAM INDHU

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ROLE: GIS Mapping and Flood-Risk Visualization

DESIGN / DEVELOPMENT DECISION:

I contributed to the GIS-based mapping and visualisation component of the disaster-management system. My focus was on displaying affected locations, flood-risk areas, evacuation zones, shelters, and important geographical information on maps. I designed the concept so that users could understand the disaster situation through visual geographic information.

CHALLENGE FACED AND SOLUTION:

The main challenge was presenting large amounts of geographical information without making the interface confusing. I addressed this by organising locations according to risk levels and using clear map markers and region-based information. This helps users quickly identify dangerous areas and nearby safe locations.

LEARNING GAINED:

I learned how Geographic Information Systems can be used in disaster-management applications. I understood that location-based information is important for identifying affected communities, planning evacuation routes, and locating emergency shelters.

SDG RELEVANCE:

The project supports SDG 11 by improving community safety and disaster resilience. GIS-based visualisation can help authorities understand affected regions and make better evacuation decisions.

COURSE OUTCOME CONTRIBUTION:

My contribution supports course outcomes related to system design, data visualisation, user requirements, interface development, and software integration. I learned how geographical information can be integrated into a software application to support real-world decision-making.

POSSIBLE IMPROVEMENT:

I would improve the GIS module by adding live satellite information, detailed flood-depth maps, dynamic risk zones, offline maps, location tracking, and more accurate route recommendations during emergencies.